Last time: Showed if A is an m x n matrix, then $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$ defined by $T(x) = Ax$ is a linear transformation.

Now interested in the reverse, is every linear transformation between these sets of the type Ax for some matrix A of the stated size? Before we that, go over a couple of useful terms related to related to functions in general that might help us with this.

A generic (not necessarily linear transformation or any special properties) function $f: B \dashrightarrow C$ is <u>one-to-one</u> (<u>injective</u> is another term for it) if for any $b_1 \neq b_2$ we have $f(b_1) \neq f(b_2)$

That is, a function where any two distinct inputs must have distinct outputs.

A function is said to be <u>onto</u> (<u>surjective</u> is other term). if $f(B) = C$, i.e, codomain of f is same as its range.

Last time we talked about the difference between codomain and range, only same if function is onto.

Again do not assume those are always the same as that would imply every function is onto which is certainly not the case!

Whether a function is one-to-one and/or onto is an important question that requires investigation.

From $\mathbb{R}$ to $\mathbb{R}$, x^2, x^3, e^x, $\sin x$ are common functions. x^2, $\sin x$ are neither one-to-one or onto, x^3 is both while e^x is one-to-one but not onto.

What about $T: \mathbb{R}^3 \text{---->} \mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (x_1 - 2x_2 + 3x_3, x_1 + 5x_3, x_2 + 7x_3)$$

Is this a linear transformation? If it is how do we check if this is one-to-one and/or onto?

Maybe if expressible via matrix that will give us a better idea on how to assess these issues? Hence why we want to show all such linear transformations can be expressed via matrices (i.e. as <u>matrix transformation</u>).

Consider these "standard" vectors in $\mathbb{R}^n$, later called standard basis when we expand on this in Chapter 4.

$$e_1 = (1, 0, 0, 0, \ldots, 0, 0)$$
$$e_2 = (0, 1, 0, 0, \ldots, 0, 0)$$
$$\vdots$$
$$e_j = (0, 0, \ldots, 0, 1, 0, \ldots, 0, 0)$$
$$\vdots$$
$$e_n = (0, 0, \ldots, 0, 0, 0, \ldots, 0, 1)$$

j-th position

That is, for $1 \leq j \leq n$, e_j has the j-th position as 1, all other positions as 0.

These vectors span $\mathbb{R}^n$ as how to write any vector in $\mathbb{R}^n$ as a linear combination of these is quite clear or can note that the matrix with these e_j's as columns has 1 as every diagonal entry, 0s elsewhere so does have a pivot in every row as needed.

Big key is, for any linear transformation $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$ there is an $m \times n$ matrix A for which $T(x) = Ax$ for all x and it turns out that A is:

$$A = \left[\underbrace{T(e_1)\ T(e_2)\ \ldots\ T(e_n)}_{\text{each of these being an } m \times 1 \text{ column}} \right]$$

Why? Well will show a specific example, general case is same idea, just a bit clumsier notationally.

<u>Ex:</u> Say $f: \mathbb{R}^2 \dashrightarrow \mathbb{R}^3$ is a linear transformation satisfying

$$f(1, 0) = (1, 2, 3)$$
$$f(0, 1) = (4, 6, 9)$$

What is f(5, 7)? First compute via linear combination

$$\begin{aligned} f(5, 7) &= f[5(1, 0) + 7(0, 1)] \\ &= 5\, f(1, 0) + 7\, f(0, 1) \end{aligned}$$

because f is linear transformation

$$\begin{aligned} &= 5\,(1, 2, 3) + 7\,(4, 6, 9) \\ &= (33, 52, 78) \end{aligned}$$

Suppose we compute this via suggested matrix. Here

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 6 \\ 3 & 9 \end{bmatrix}$$

Then

$$A \begin{bmatrix} 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 2 & 6 \\ 3 & 9 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 33 \\ 52 \\ 78 \end{bmatrix}$$

Same as earlier. Key is that matrix calculation is equivalent to the linear combination of the columns. And that's why this works in general, same idea, notation clumsier, but idea identical.

So any linear transformation $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$ can be represented in the form $T(x) = Ax$ for A some fixed $m \times n$ matrix.

Side note that won't really be relevant until later in course: This only works for $\mathbb{R}^n$ to $\mathbb{R}^m$, in more general settings it may not be true.

Although a bit early for this, will give a brief idea of an example which illustrates this issue.

Say our domain is a space of differentiable functions. Consider the transformation:

$$T(f) = f'$$

This will be linear since derivative behaves well under sum and constant multiple. But realistically this can't be represented as a matrix, what matrix could map $\sin x$ to $\cos x$ while also taking e^x to e^x amongst the myriad of other things needed? So this can't be represented as a matrix transformation.

Book delves heavily into some of the common linear transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$, the matrices which represent them, etc.

Look those over to get an idea how they work, but don't spend your time memorizing them - not super useful, too specific a context, more helpful to think in broader contexts.

How to tell if a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is one-to-one? Again this is broader than the specific transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Before this, how to tell if some generic f: B ----> C is one-to-one? The typical thing to try is to set

$$f(b_1) = f(b_2)$$

and see if this forces $b_1 = b_2$. If so then f is one-to-one.

Why does that fail with say $f(x) = x^2$ on $\mathbb{R}$?
Well trying this there you get:

$$b_1^2 = b_2^2$$

----> $$(b_1 - b_2)(b_1 + b_2) = 0$$

So $b_1 = b_2$ is not forced, $b_1 = -b_2$ is also possible which is why this is not one-to-one.

In contrast $g(x) = x^3$, $b_1^3 = b_2^3$ cube rooting both sides gives $b_1 = b_2$, safe since odd powers have no issues with negatives so $g(x)$ is one-to-one.

Again, especially from $\mathbb{R}$ to $\mathbb{R}$ this is a common strategy to prove one-to-one, especially because if it happens not to be one-to-one, you will tend to run into trouble when you try to prove it is (when of course it is not), and that will often be suggestive of an example of why it is not (say a pair of inputs which have the same output). That's essentially what happened with x^2.

Can we try this idea to see if some specific linear transformation is one-to-one? Perhaps but in one sense seems like it could be much harder since you are going from $\mathbb{R}^n$ to $\mathbb{R}^m$ which is obviously tougher than $\mathbb{R}$ to $\mathbb{R}$. On the other hand, linear transformations are very special functions so that makes it likely we can do it more easily.

In short, due to those issues, it probably will work quite differently in this setting.

For such a linear transformation $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$ T being one-to-one is equivalent to T not having having a nontrivial solution to $T(x) = 0$ is a nice thing. Why?

Well if T has such a nontrivial solution, of course T cannot be one-to-one as T maps both 0 and that nontrivial solution to 0, immediately defying the definition of one-to-one.

On other hand, if T has no such nontrivial solution, then T must be one-to-one for if we had $x_1 \neq x_2$ where $T(x_1) = T(x_2)$, then via linear transformation $T(x_1 - x_2) = 0$ would give us a nontrivial solution to $T(x) = 0$.

With this in mind realize that, if A is the associated matrix for T, checking $T(x) = 0$ is the same as checking $Ax = 0$.

And based on earlier ideas that boils down to whether or not A has a nonpivot column which we know how to check.

How to check if same type of T is onto or not?

If it is, then $\forall\ b \in \mathbb{R}^m$, $T(x) = b$ must have a solution.

And this means we need $Ax = b$ to have a solution for all such b. And again from earlier ideas, that happens exactly when A has a pivot in every row.

For earlier example:

$$T(x_1, x_2, x_3) = (x_1 - 2x_2 + 3x_3, x_1 + 5x_3, x_2 + 7x_3)$$

via plugging in the three standard vectors we find the matrix for T to be

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 0 & 5 \\ 0 & 1 & 7 \end{bmatrix}$$

So checking REF of A we can determine if T is one-to-one and/or onto.

1.9.44 Say $S: \mathbb{R}^p \dashrightarrow \mathbb{R}^n$, $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$

are linear transformations. Is $T \circ S: \mathbb{R}^p \dashrightarrow \mathbb{R}^m$ a linear transformation?

This is composition of functions so important that range of first applied function be contained in domain of second or this won't make sense. For instance $S \circ T$ won't be defined if $p \neq m$

For any $x, y \in \mathbb{R}^p$, $c \in \mathbb{R}$

$$
\begin{aligned}
(T \circ S)(cx + y) &= T[S(cx + y)] \\
&= T[cS(x) + S(y)] \quad \text{since } S \text{ is linear trans.} \\
&= cT(S(x)) + T(S(y)) \quad \text{since } T \text{ is lin. trans.} \\
&= c(T \circ S)(x) + (T \circ S)(y) \quad \text{since } T \text{ is lin. trans.}
\end{aligned}
$$

This shows that $T \circ S$ is a linear transformation.
So that being the case how would we compute its associated matrix?

Say S's matrix is B and T's is A. Then

$$
\begin{aligned}
(T \circ S)(x) = T(S(x)) &= T(Bx) \\
&= A(Bx) \\
&\overset{?}{=} (AB)\,x
\end{aligned}
$$

Last statement unclear at moment. But either way seems we need to "multiply" A and B. But what does that mean? We defined multiplication of matrices only for Ax, A being m x n and x being n x 1, x was restricted to only one column, must generalize that.

Why more than one column needed? Well could be many reasons, but one of them could relate to an issue we've been tackling recently - how to check if a vector is in the span of a set of vectors. We did this by checking the augmented matrix, with that vector being the augmented column.

So we check if that system is consistent.

But what if we wanted to check multiple vectors as to whether or not they are in the span in question. Well we don't want to row reduce a bunch of different matrices.

Could maybe we do it just by using additional extra columns at once? If so that might be one possible use of dealing with multiple extra columns.

Also, we are suggesting "multiplying" two matrices to create the composition we are after.

But say $f, g: \mathbb{R} \longrightarrow \mathbb{R}$. Aren't the product $f \cdot g$ and the composition $f \circ g$ often very different functions?

Indeed they are! So why would "multiplication" of matrices be associated with composition of linear transformations? Lot of issues to clarify here.

This leads well into the next chapter.

Chapter 2: Matrix Algebra

2.1 Matrix Operations

Want to clarify various things about working with matrices, mostly addition and multiplication.

For instance if we wanted to add two linear transformations maybe addition of matrices? And as mentioned, to compose them, maybe multiplication?

Addition is natural and most have no issues with that. Multiplication much harder both in definition and in application so have to do that more carefully.

To get us started, recall standard notation which we already have mentioned

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Again a_{ij} means entry in i-th row and j-th column. Should be sure to be familiar with that notation as often when we refer to a matrix A we will use a_{ij} for that entry without clarifying since we don't want to have to write out the matrix A in full each time, that's just annoying.

The entries $a_{11}, a_{22}, \ldots, a_{kk}, \ldots$ are the <u>diagonal</u> of the matrix (mostly used in square matrices).

Recall in square matrix, upper triangular means all entries below diagonal are 0 while lower triangular means all above diagonal are 0. What if matrix is both?

That means all entries not on diagonal are 0. This is another type of important matrix, especially in matrix multiplication, called a <u>diagonal matrix</u>, which again is a square matrix where every off diagonal entry is 0.

By nature those are much easier to work with than generic matrices, hence the motivation, you will see things related to those throughout this class as a result.